

ABHILASH NEOG

🌐 [abhilash-neog.github.io](https://github.com/abhilash-neog) [in](#) LinkedIn [🎓](#) Scholar [G](#) Github [✉](mailto:abhilash22@vt.edu) abhilash22@vt.edu

Research Summary

Ph.D. researcher in machine learning with experience developing foundation models, generative models, and production ML systems. Work spans model architecture and multi-GPU training to ML pipelines and deployment, with recent research in diffusion and flow models, multimodal learning, and time-series foundation models.

Education

Virginia Tech

Ph.D. in Computer Science

M.S. in Computer Science

Advisor: *Dr. Anuj Karpatne* | GPA: 4.0/4.0

Blacksburg, VA

Aug 2022 – May 2027

Aug 2022 – Dec 2024

Birla Institute of Technology and Science (BITS), Pilani

Bachelor of Engineering (B.E.) in Computer Science, GPA: 8.08/10

Pilani, India

July 2016 – July 2020

Work Experience

Applied Scientist Intern | [Microsoft \(MAI\)](#)

May 2026 – Aug 2026

- Designed and trained a FLUX-based generative image model using LoRA fine-tuning and adaptive wavelet supervision, improving FID by 7.7% over state-of-the-art image-outpainting baselines.
- Built a 12K-image high-quality subject-clarity dataset and automated data-curation pipeline using GPT-5.5 and prompt-based segmentation models.
- Developed an end-to-end image-generation pipeline integrating VLM-based semantic guidance, subject segmentation, high-frequency feature estimation, and additional image conditioning, while retaining the inference cost and latency of the base FLUX model; resulted in a patent submission.

Machine Learning Intern | [Kryptowire](#)

May 2023 – Aug 2023

- Developed an outlier-detection model to denoise sensor-based Human Activity Recognition (HAR) time-series data
- Built and deployed a Keras CNN for HAR to an Android application using TensorFlow Lite, achieving an 82% F1 score

Data Scientist | [Oracle](#)

Sep 2020 – July 2022

- Built and deployed production ML applications within ETL pipelines using Spark, MLOps, & CI/CD infrastructure.
- Designed and deployed a DeepAR-based demand-forecasting system spanning data processing, model training, evaluation, and production integration
- Developed an unsupervised classification system for a 71k-label taxonomy, improving accuracy by 40% over prior LLM baselines

Software Engineer Intern | [VMware](#)

Jan 2020 – June 2020

- Streamlined Workspace ONE Cloud data workflows using Spring Boot REST APIs, and contributed to an end-user federation application with JUnit/Mockito testing

Software Engineer Intern | [Samsung Research Institute](#)

May 2019 – July 2019

- Optimized the C++ HAL call flow for Live Focus under low-light conditions, achieving considerable noise reduction during the first phase of camera capture

Research Experience

Graduate Research Assistant | [Knowledge-guided Machine Learning Lab](#)

Jan 2023 – Present

- Working on cross-modal alignment between LLMs and time-series foundation models using diffusion transformers and flow matching for knowledge-conditioned generation
- Designed and trained LakeFM, a foundation model for irregular multivariate time series, with custom tokenization and query-based forecasting, and multi-GPU training, achieving 22% lower average MSE error than the strongest competing foundation models
- Contributed to a physics-guided Diffusion Model for solving Partial Differential Eqns., achieving $15-250\times$ faster inference than SOTA baselines
- Vision-Language Model (VLM) evaluation for visual reasoning, fine-grained attribute grounding and zero-shot scene graph generation; contributed to fine-grained segmentation dataset generation and segmentation model benchmarking
- Designed and developed Model-agnostic Adapters for Irregular and Sparse Time-Series modeling

Selected Publications (*denotes equal contribution)

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen, Jian Jiao. “Preserving Subject-Clarity in Image Outpainting with Multiscale Wavelet Supervision”. [Under Review](#)
2. **Abhilash Neog**, Sepideh Fatemi, Medha Sawhney, et al. “LakeFM: Toward a Foundation Model for Aquatic Ecosystems Using Irregular Multivariate Multi-depth Time Series Data”. [KDD 2026](#)
3. **Abhilash Neog**, Arka Daw, Sepideh Fatemi, Medha Sawhney, et al. “Investigating a Model-Agnostic and Imputation-Free Approach for Irregularly-Sampled Multivariate Time-Series Modeling”. [TMLR 2026](#)
4. A. Pandey*, **Abhilash Neog***, Gautam Jajoo. “On the Internal Semantics of Time-Series Foundation Models”. [BERT2S @ NeurIPS 2025](#)
5. M. Sawhney, **Abhilash Neog**, Mridul Khurana, Anuj Karpatne. “PRISMA: Improving the Accuracy–Latency Frontier of Diffusion-based PDE Solvers Using Physics-Informed Spectral Attention”. [Under Review](#)
6. S. Fatemi, **Abhilash Neog**, Amartya Dutta, M. Sawhney, et al. “Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance”. [\(Oral\) CV4Science @ CVPR 2025](#)
7. A. Dutta, M. Sawhney, K.S. Mehrab, **Abhilash Neog**, Mridul Khurana, et al. “Open World Scene Graph Generation using Vision Language Models”. [CVinWild @ CVPR 2025](#)
8. KS Mehrab, M. Maruf, Arka Daw, **Abhilash Neog**, HB Manogaran, et al. “Fish-Vista: A Multi-Purpose Dataset for Understanding & Identification of Traits from Images”. [CVPR 2025](#)
9. M. Maruf, Arka Daw, KS Mehrab, HB Manogaran, **Abhilash Neog**, M. Sawhney, et al. “VLM4Bio: A Benchmark Dataset to Evaluate Pretrained Vision-Language Models for Trait Discovery from Biological Images”. [NeurIPS 2024](#)

Patents

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen. [Microsoft, 2026](#). “Subject-Clarity-Aware Image Outpainting via Semantic and Structural Guidance with Vision-Language Models and High-Frequency Map Estimation.” [Patent submission \(Under Review\)](#).
2. Akash B., Krishnan R., **Abhilash, N.**, Dipawesh P. and Karthik B. [Oracle, 2024](#). “Machine Learning Based Spend Classification.” [U.S. Patent Application 17/903,161](#).

Talks/Awards

- Invited talk at the [ASLO 2026 Conference - Session on AI for the Next Frontier in Aquatic Sciences](#)
- Invited talk at the [AGU 2025 Conference - Session on Advancing Water Science Through AI](#)
- [NSF NAIRR \(National AI Research Resource\) Pilot Award, 2024](#)
- “*Star of the Month (Dec 2021)*” within the Oracle Analytics Cloud Organization, Oracle India

Academic Duties

- **Reviewing:**
 - Conferences/Journal - NeurIPS ('25, '26), AAAI ('26, '27), TMLR, ICML 2026, KDD 2026, ECCV 2026
 - Workshops - FM4Science (ICLR '26), FMSD (ICML '25, '26), FMs in the Wild (ICLR '25), Open Science for FMs (ICLR '25), XAI4Science (ICLR '25), ML4PS (NeurIPS '25), TSALM (NeurIPS '24)
- **Teaching:**
 - Graduate Teaching Assistant, CS 5805 Machine Learning, Spring 2024
 - Teaching Assistant, BITS F312 Neural Networks and Fuzzy Logic, Fall 2019

Technical Skills

Languages: Python, C++, Java, SQL, PySpark

ML: PyTorch, TensorFlow Lite, Keras, Huggingface Transformers, Diffusers, vLLM

Software: REST APIs, Spring Boot, Git, Hydra, JUnit/Mockito